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Experiment No: 01.

Aim: Study of plane table and accessories.

Objective: Study of plane table by traversing method.

Introduction to plane table: Plane table surveying is a graphical method of surveying in which field work and plotting are done simultaneously in the field. The plane table consists of the following:-

- Drawing board mounted on tripod.
- Straight edge called alidade.

The drawing board: The board is made of well seasoned wood and varies in size from 40cm X 30cm to 75cm X 60 cm or 50cm X 60 cm square.

The Alidade: The alidade consists of metals or boxwood straight edge or ruler about 50 cm long. The edge of the alidade is called the fiducially edge.

Accessories to the plane table:

- Through Compass: The compass is used to mark the direction of the meridian on the paper.

- U-frame or Plumbing fork: U-frame or plumb

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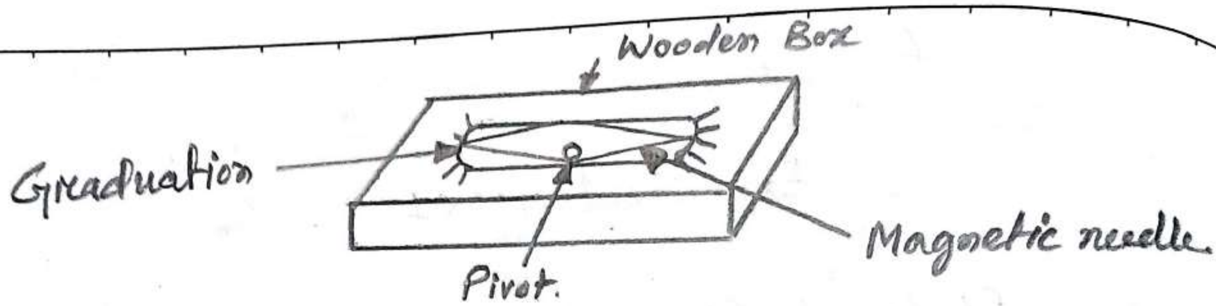


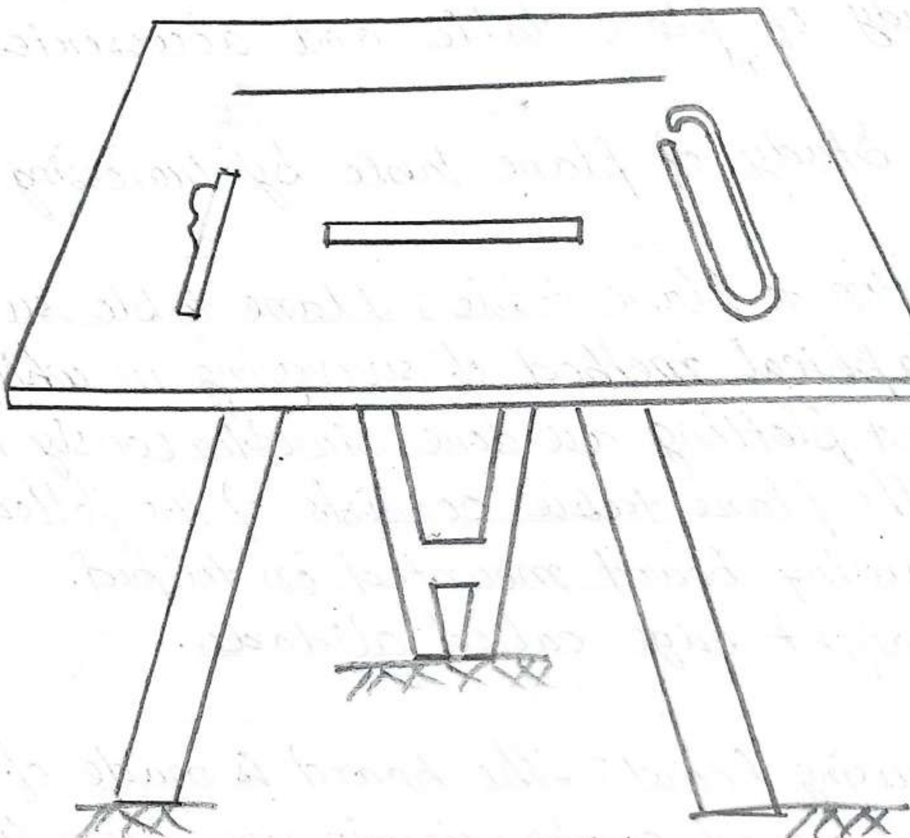
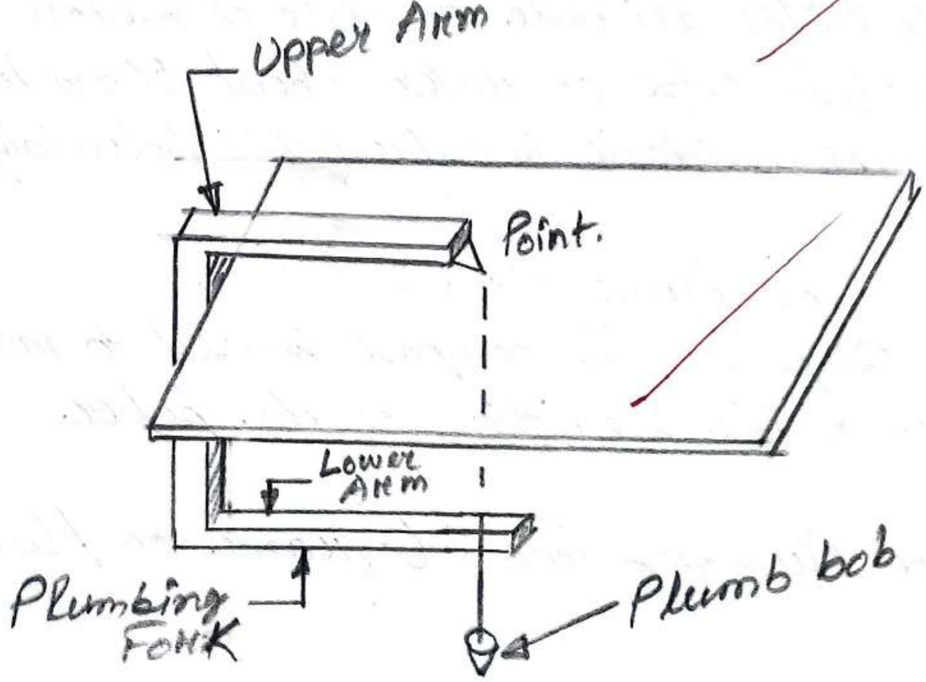


Fig: Plane Table



bob used for centering the table.

⊛ Water proof cover : Water proof cover protects the steel from rain.

⊛ Spirit level or level tube : A level tube is used to level the plane table.

⊛ Drawing sheet : The drawing sheet is fixed on the top of the drawing board.

⊛ Pencil and eraser : A pencil is used for constructing lines and eraser is used for erasing lines after completion of the plan.

Setting up the Plane table : The setting up the plane table includes the following three operations :

(i) Centering of the plane table.
(ii) Levelling the plane table.
(iii) Orientation of plane table.

(i) Centering the plane table : The table should be set up at a convenient height for working say about 1m. The legs of the tripod should be spread well apart and firmly fixed on to the ground. Then the operation of centering is carried out by means of U-frame and plumb bob. The plane table is exactly placed over the ground station by U-frame and plumb bob.

(ii) Levelling the Plane table: The process of levelling is carried out with the help of level tube. The bulstate of level tube is brought to centre in two directions, which are right angles to each other. This is achieved by moving legs.

(iii) Orientating the table: The process of keeping the plane table always parallel to the position, which is occupied at the first station, is known as orientation. When the plane table is oriented, the lines on the board are parallel to the lines on the ground.

The traversing Method: This method is suitable for connecting the traverses station. This is similar to compass traversing or theodolite traversing. But here fielding and plotting are done simultaneously with the help of the radiation and intersection methods.

Procedure:

(a) Suppose A, B, C and D are the traverse stations.

(b) The table is set up at the station A. A suitable point a is selected on the sheet is such a way that the whole area may be plotted on the sheet. The table is centred, levelled and clamped. The north line marked on the right hand top corner of the sheet.

(c) With the alidade touching touching point a the ranging rod at B is bisected and a ray is down. The distance AB is measured and plotted to any suitable scale..

(d) The table is shifted and centred over B. It is then levelled, orientated by back-sighting and clamped.

(e) With the alidade touching point b, the ranging rod at C is bisected and a ray is down. The distance BC is measured and plotted to the same scale.

(f) The table is shifted and set up at C and the same procedure is repeated.

(g) In this manner, all stations of the traverse are connected.

(h) At the end, the finishing point may not coincide with the starting point and there may be some closing error. This errore is adjusted graphically by Bouditch's rule.

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Experiment No: 02.

Aim: Measurement of horizontal angles theodolite by method of repetition (repetition).

Apparatus: Theodolite, three ranging rods.

Theory: Theodolite is an instrument designed for the measurement of horizontal and vertical angle. It is most precise method it is also used for laying of horizontal angles. Locating points on the line prolonging the survey line establishing the gradient, determination of difference in the elevation setting out curve. Theodolite are of two types transit and non-transit. Transit theodolite is commonly used now-a-days. In transit theodolite, telescope can be revolved a complete revolution about its horizontal axis in a vertical plane.

Repetition method of measuring horizontal angles:

When it is required to measure horizontal angles with great accuracy as in the case of traverse, the method of repetition may be adopted. In this method the same angle is added several times by keeping the vernier to remain clamped each time at the end of each measurement instead of setting it back to zero when sighting at the previous station. The corrected horizontal angle is then obtained by dividing the final reading by the number of repetitions. Usually six reading, three with

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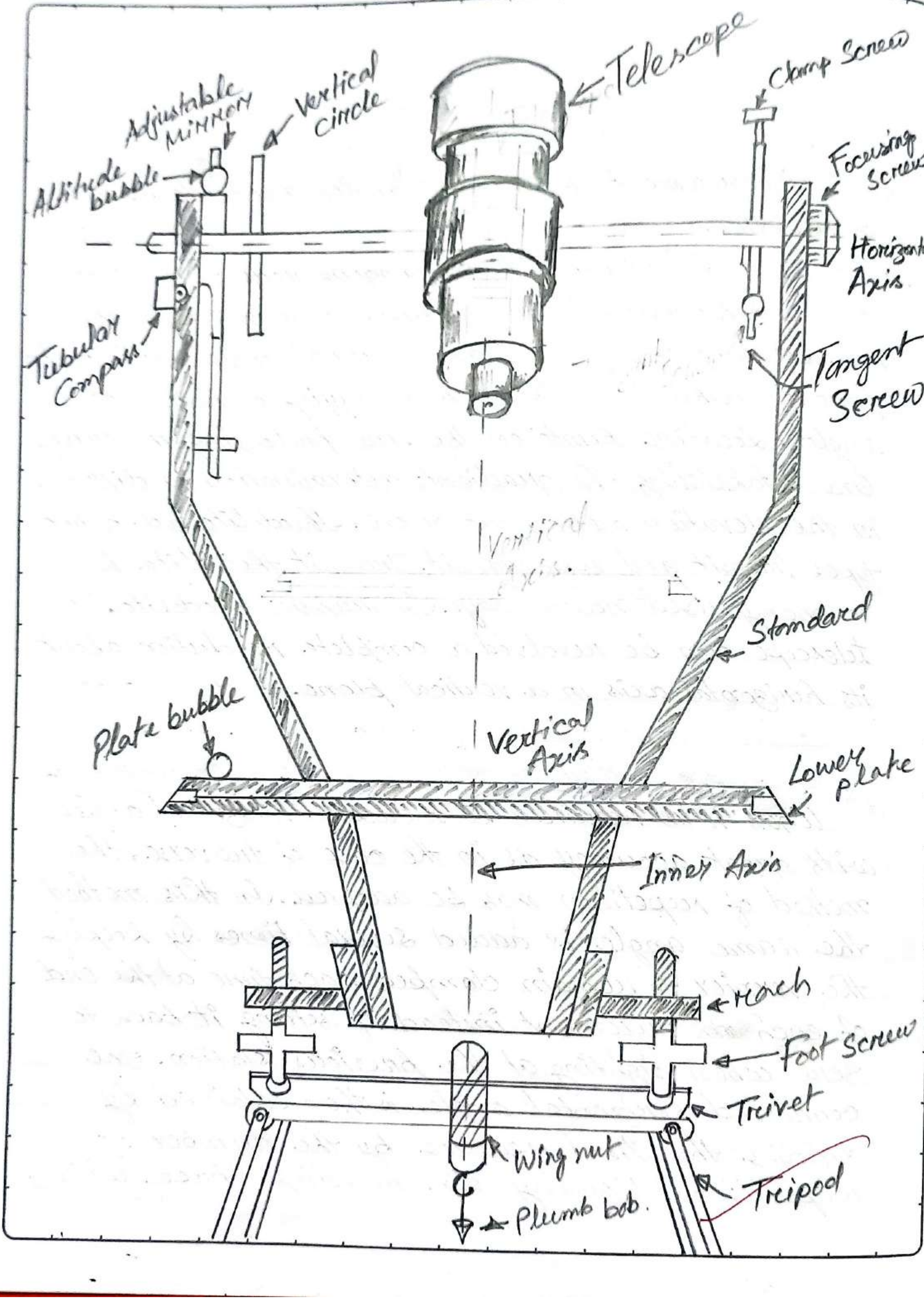
Telescope
Clamp Screw
Focusing Screw
Horizontal Axis
Tangent Screw
Adjustable Mirror
Vertical circle
Altitude bubble
Tubular Compass
Vertical Axis
Standard
Plate bubble
Vertical Axis
Lower plate
Inner Axis
Trivet
Foot Screw
Trivet
Wing nut
Tripod
Plumb bob.

face left and three with face right, are taken. The average horizontal angle is then calculated.

Procedure:

(i) Let ∠LOM is the horizontal angle to be measured. O is the station point fixed on the ground by a peg 'O' and level it accurately.

(ii) Set the horizontal graduated circle vernier A to read 0 or 360° by copper clamp screw and slow motion screw. Clamp the telescope to bisect the bottom shoe of the flag fixed at point 'L' and tighten the lower clamp. Exactly intersect the centre of the Survey-I button shoe by means of lower slow motion screw. Check that the face of the theodolite should be left and the telescope in normal position.

(iii) Check the reading of vernier A to see that no slip has occured. Also see that the plate levels are in the centre of their run. Read the vernier B also.

(iv) Release the upper clamp screw and turn the theodolite clockwise. Bisect the flag bottom shoe fixed at point M by a telescope. Tighten the upper clamp screw and bisect the shoe exactly by means of upper slow motion screw.

(v) Note the reading on both of the vernier to get the approximate value of the angle ∠LOM.

(vi) Release the lower clamp screw and rotate the theodolite anticlockwise. Bisect again the bottom shoe of the flag ~~fixed~~ at 'L' and tighten the lower clamp screw. By means of slow motion screw bisect exactly the centre of the shoe.

(vii) Release now the upper clamp screw and rotate the theodolite clockwise. Bisect the bottom shoe of the flag fixed at 'M' and tighten the upper clamp screw. By means of slow motion screw bisect exactly the centre of the shoe. The vernier readings will be now twice the angles.

(viii) Repeat the process untill the angle is repeated the required number of times (usually 3). Add 360° to every revalution to the final reading and divide the total angle by number of repetitions to get value of angle ∠LOM.

(IX) Change the face of the theodolite the telescope will now be inverted. Repeat the whole process exactly in the above manner and obtain value of angle ∠LOM.

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(X) The average horizontal angle is then obtained by taking the average of the two angles obtained with face left and face right.

(XI) Usually three repetitions face left and three with face right should be taken and the mean angle should be calculated.

Observation:

Station	obs.	Face	No. of reading	Initial reading on vernier		Final reading on vernier		Angle of vernier		Mean angle on vernier.	Mean angle of obs (hz angle)
				A	B	A	B	A	B		
P	2	left	2	0°0'	180°0'	68°50'	248°48'	137°5'	137°7'	137°6'	
				68°50'	248°48'	137°5'	317°7'				136°59'
P	2	Right	2	0°0'	180°0'	69°39'	248°40'	136°5'	136°52'	136°52'	
				69°39'	248°40'	136°52'	316°56'				

Station	obs.	Face	No. of reading	Initial reading on vernier		Final reading on vernier		Angle on vernier		Mean angle on vernier	Mean angle of obs.
				A	B	A	B	A	B		
P	2	left	2	0°0'	180°0'	68°50'	248°48'	137°5'	137°7'	137°7'	
				68°50'	248°48'	137°5'	317°7'				136°59'
P	2	Right	2	0°0'	180°0'	69°39'	248°40'	136°5'	136°5'	136°52'	
				69°39'	248°40	136°52'	316°56'				

Teacher's Signature:

Experiment NO: 03.

Aim: Measurement of vertical angle with theodolite.

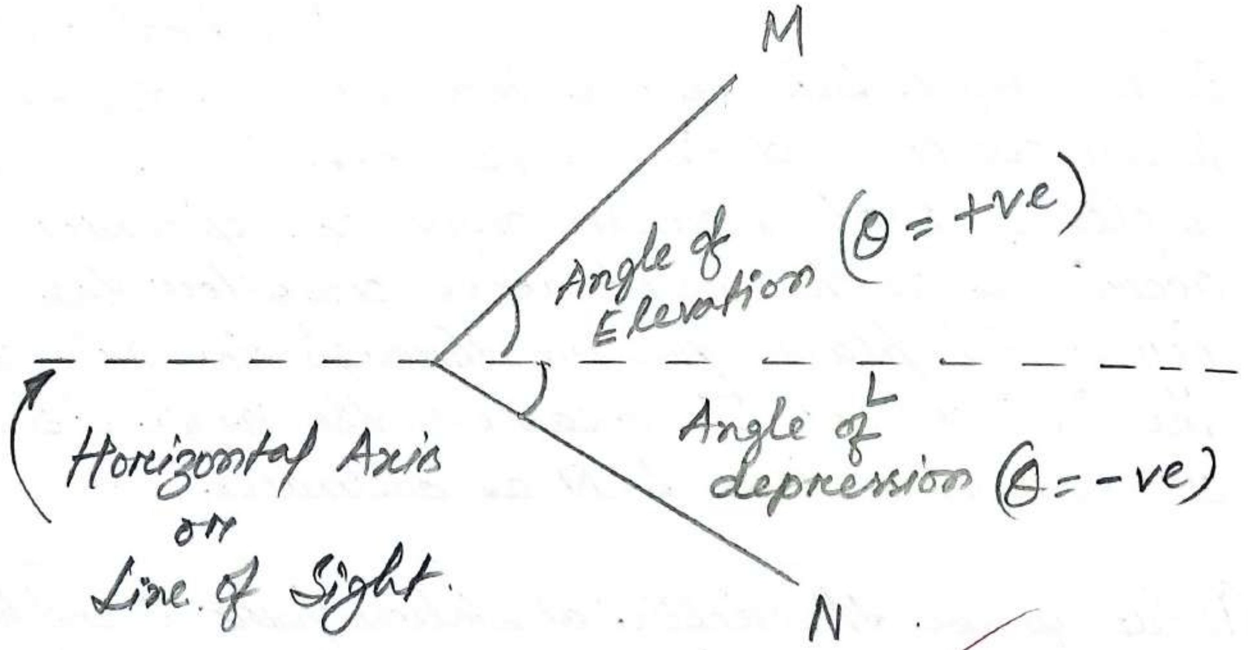
Apparatus: Theodolite.

Measurement of vertical angle: A vertical angle is the angle between the inclined line of sight to an object and the horizontal. It may be an angle of elevation or an angle of depression according as the point above or below the horizontal plane passing through the axis of the instrument. To measeure the angle of elevation or depression, take LOM as ordinates.

(i) Set up the theodolite at station point O and level it accurately with reference to the altitude level.

(ii) Set vertical vernier C and D exactly to zero by using the vertical circle clamp and tangent screw, while the altitude level should remain in the centre of its run. Also the face of the theodolite should be left.

(iii) Release the vertical circle clamp screw and rotate the telescope in vertical plane so as to


M
Angle of Elevation (θ = +ve)
Angle of depression (θ = -ve)
Horizontal Axis
or
Line of Sight
N

bisect the object M. Tighten the vertical circle clamp and exactly bisect the object by slow motion screw.

(iv) Read both vernier C and D, the mean of the two readings gives the value of the required angle.

(v) Similar observation may be made with other face. The average of the values thus obtained gives the value of the required angle which is free from instrumental errors.

(vi) Similarly the angle of depression can be measured following the above steps.

Calculation: (For angle of elevation)

Station	Object	Face	Readin on vernier		Angular vernier		Mean Angle. on vernier	Mean angle of observation (θ_1)
			C	D	C	D		
P	Horizontal	Left	0°0'0"	0°0'0"	1°31'40"	1°31'40"	1°31'40"	
	Q	Pt(I)	1°31'40"	1°31'40"				
		Pt(II)	—	—				
		Pt(III)	—	-				
P	Horizontal	Right	0°0'0"	0°0'0"	5°53'40"	5°53'40"	5°53'40"	
	Q	Pt(I)	5°53'40"	5°53'40"				
		Pt(II)	—	—				
		Pt(III)	—	-				

For angle of depression:

Station	Object	Face	Reading on vernier C	Reading on vernier D	Angle on vernier C	Angle on vernier D	Mean angle of Vernier	Mean angle of observation (θ_2)
P	Horizontal	Left	0°0'0"	0°0'0"	5°37'20"	5°37'20"	5°37'20"	
	R	Pt.(I)	5°37'20"	5°37'20"				
		Pt.(II)						
		Pt.(III)						
P	Horizontal	Right	0°0'0"	0°0'0"	1°25'20"	1°25'20"	1°25'20"	3°31'20"
	R	Pt.(I)	~~5°37'20"~~	~~5°37'20"~~				
		Pt.(II)	1°25'20"	1°25'20"				
		Pt.(III)						

D = Distance = 25.6 m.

$\therefore h_1 = D \tan\theta = 25.6 \times \tan(3°42'40'')$
$= 1.66$ m

$h_2 = D\tan\theta = 25.6 \times \tan(3°31'20'')$
$= 1.57$ m

$\therefore$ Height $= h_1 + h_2 = 1.66 + 1.57$ m.
$= \underline{\underline{3.23 \text{ m}}}$

Experiment No: 04.

Aim: Setting out of simple circular curve by Rankine method of tangential angle.

Apparatus: Theodolite, ranging rods, pegs, flags, taper.

Objective: Two tangent intersect at chainage (180+10) the deflection angle being 20°. Calculate all the data necessary for setting out a 3° simple circular curve by method of deflection angle (tangential angle). The peg interval may be taken as 30 m.

Procedure for setting out curve:

(1). Locate the tangent points T_1 and T_2 on the straights AB and CB.

(2). Set up the theodolite at the beginning of the curve T_1.

(3). With the vernier A of the horizontal circle set to zero, direct the telescope of the ranging rod fixed at the point of intersection B and bisect it.

(4). Unclamped the vernier plate and set the vernier A to the first tangential angle α_1, the telescope being thus directed along U.D.

(5). Measure along the line TLD, the length equal to first subchord (C_1) thus fixing first point D on the curve.

(6). Unclamped the vernier plate now and set the vernier A to the second total tangential angle α_2, the line of sight now directed along TIE.

(7). With the zero end of chain or tape at D, and with a arrow at distances of DIE = C_2 (Second Chord or say normal chord), swing the chain about D, untill the line of sight bisects the arrow, thus fixing the second point E on the curve.

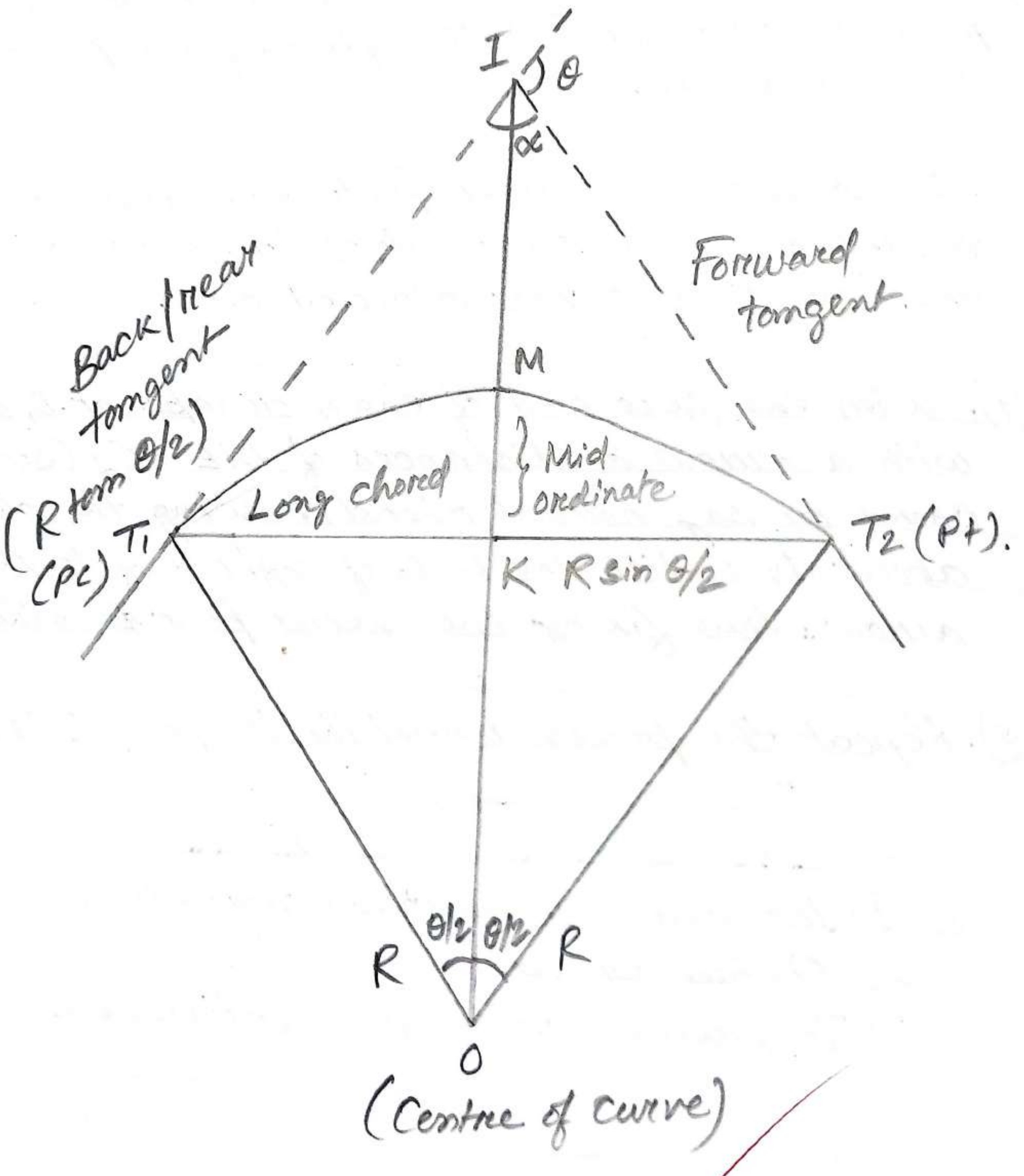
(8). Repeat the process untill the last point T_2 is reached.

* Why curve is constructed on Road?

Ans: (1) For allowing smoother transition.
(2) Control Speed.
(3) To prevent driver from fatigueness.

* Different types of curve.

Ans: (1) Horizontal curve.
(2) Vertical curve.
(3) Transition curve.


I
θ
α
Back/rear tangent
(R tan θ/2)
Forward tangent
M
Mid ordinate
Long chord
(PC) T1
T2 (PT).
K
R sin θ/2
θ/2
θ/2
R
R
O
(Centre of curve)

* Horizontal curve : (1) Simple circular curve.
(2) Compound curve.
(3) S- curve.

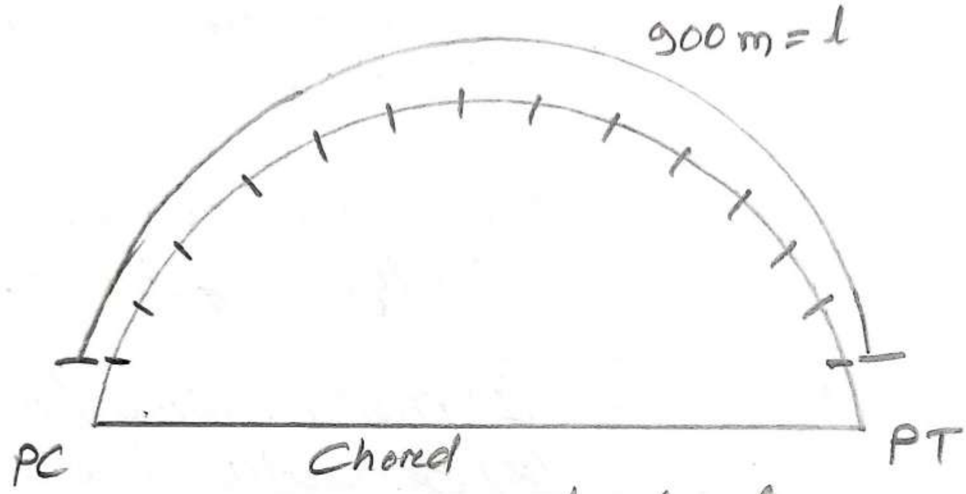
* Vertical curve : (1) Summit curve.
(2) Valley / Sag curve.

* Transition curve : (1) Cubid Spiral / clothoid.
(2) Bernoullis lexni Scale.
(3) Cubic parabola.

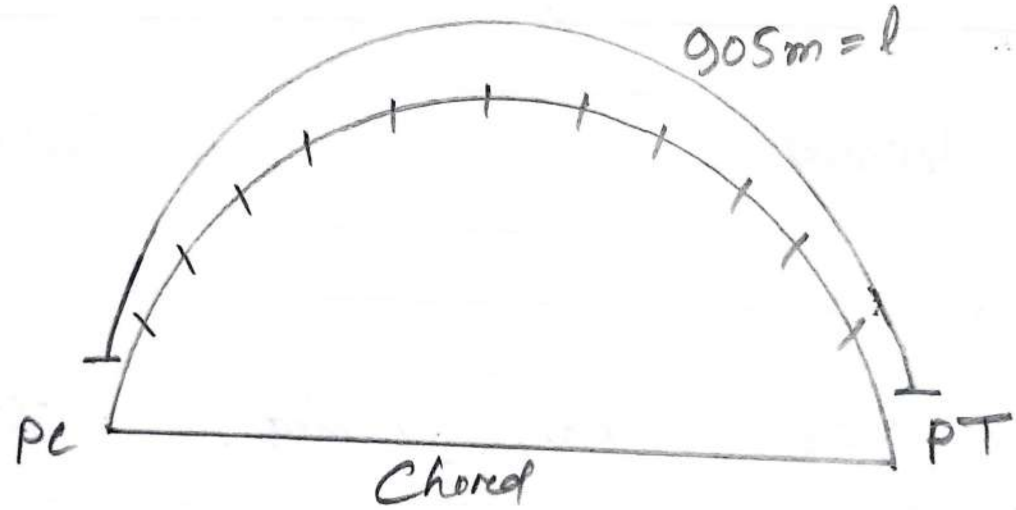
Setting out of Curve

- Linear Method
- Angular Method
  - Two theodolite Method
  - Rankine method or One theodolite method
  - Tachiometere method.

Setting out of curve : This means locating various points along the length of curve.
H circular curve.
PC → Point of curvature.
PT → Point of tangency.



eg of normal chord.



eg of Sub chord

Experiment Name / No.:

O to T_1 → Radius of curvature.
O to T_2 → Radius of curvature.
I → Intersection point.
θ → Deflection angle (angle made by rear tangent with forward tangent) is called deflection angle.
α → Intersection angle (angle made between rear and forward tangent)
$\alpha = 180° - \theta$

Long chord : Line joining between point PC & PT (minⁿ distance between PC and PT)

Then, draw a line between point I and point O.
→ Long chord will get divided into 2 equal parts.
∴ Long chord will get divided into 2 equal parts (i.e $T_1K = T_2K$)

M → Highest point of curve.
K to M → Mid ordinate / versed sine distance.
M to L → Apex distance.

(Straight line that touches a point in a curve.) → Tangent.
(Straight line / Segment that connects two points) → Chord.

Normal chord : It is the full chain length on a curve.

Sub chord : It is that distance which is shorter than the normal chord length. It usually occurs at the beginning or end of a curve.

Teacher's Signature:

Eg of ~~m~~ normal chord: refer to given diagram.

* How many times changing is requireed if we use 30m chain?

$\rightarrow \frac{900}{30} = 30$ times.

Eg. of sub-chord: reefer to given diagram.

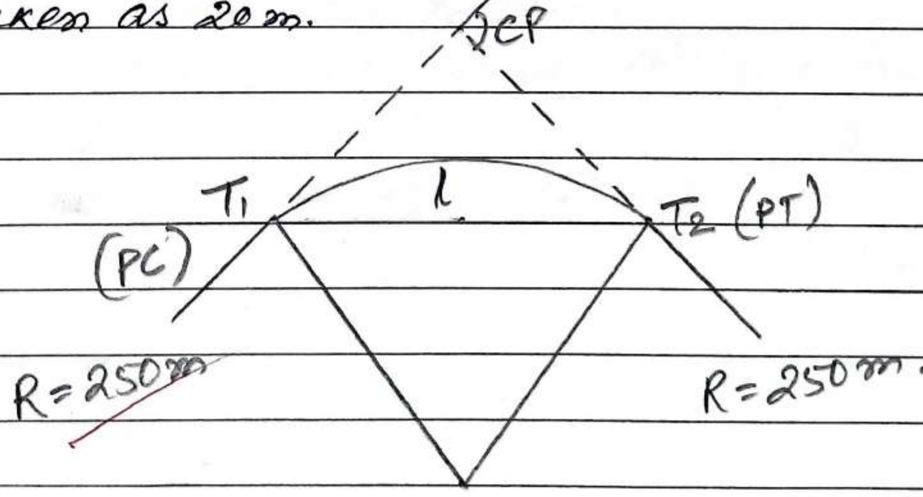
$\rightarrow \left(\frac{900}{30} + 5\right)m = 30$ times. $+ 5m$.

(Chaining)(normal chord) (Sub-chord)

Forward tangent = $R \tan \theta/2$.

Q) Two tangents intersect at chainage 1250m. The angle of intersection (α) is 150°. Calculate all data necessary for setting out a curve of 250m radius by the deflection angle method / tangential method. The peg interval may be taken as 20m.

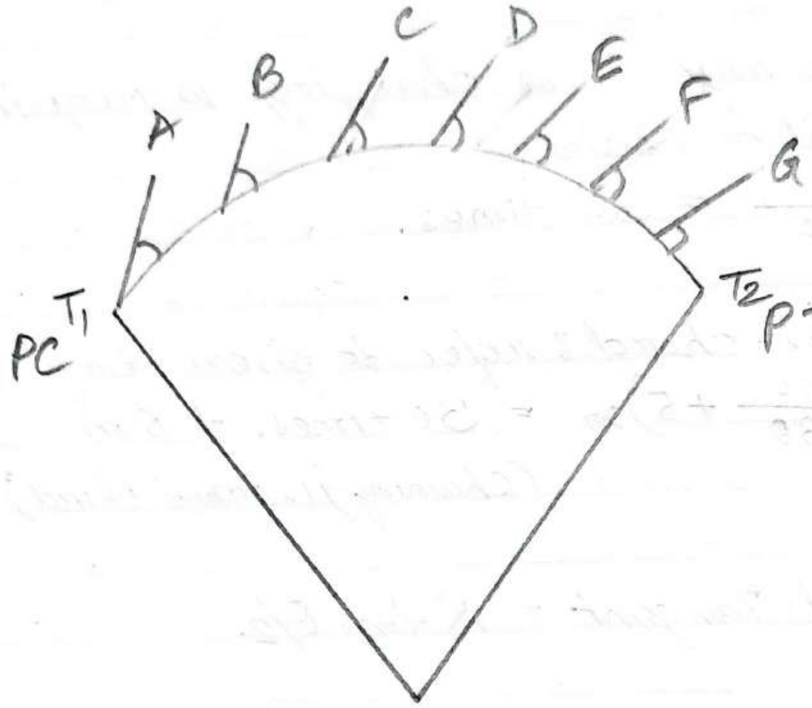
Ans:

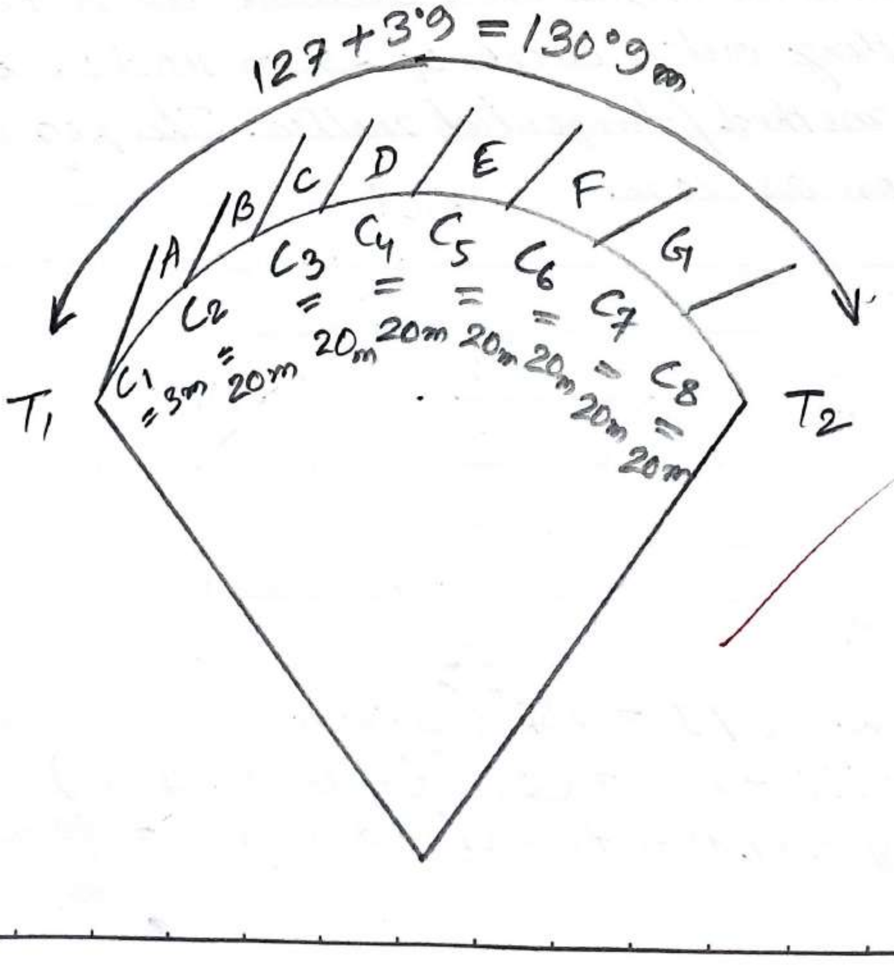


Chainage of PI = 150° (given).

$\therefore \theta = 180° - 150° = 30°$ (Deflection angle).

and Peg interval = Full chord length = 20m.


A
B
C
D
E
F
G
T1
PC
T2
PT


127 + 3·9 = 130·9 m
A
B
C
D
E
F
G
T1
T2
C1 = 3m
C2 = 20m
C3 = 20m
C4 = 20m
C5 = 20m
C6 = 20m
C7 = 20m
C8 = 20m

Steps —

(i) Tangent length (T) calculation:

$$T = R \tan \theta/2 = 250 \tan 30/2 = 67\,m.$$

(ii) Curve length (l) calculation:

$$l = \frac{\pi R \theta}{180^\circ} = \frac{\pi \times 250 \times 30}{180} = 130.9\,m$$

(iii) Chainage of PC (T_1) calculation:

Chainage of P.I − T

$$= 1250 - 67 = 1183\,m.$$

(iv) Chainage of P.T (T_2) Calculation:

Chainage of PT + l.

$$= 1183 + 130.9 = 1313.9\,m.$$

(v) Determine no. of chords:

Length of initial Sub-chord (C_1): $C_1 = 1190 - 1183 = 7\,m.$

We know,

Full chord length = 20 m (given)

$\therefore$ No. of full chords $= l/L = \frac{130.9}{20} = 6.5$

We know, length of curve = length of chord.

Hence, total length of full chord $= 6 \times 20 = 120\,m.$

The length of final Sub-chord (C_s):–

$$C_s = 130.9 - 127 = 3.9\,m.$$

∴ There are total 8 nos of chords where,

$C_1 = 7m$.

$C_2 = C_3 = C_4 = C_5 = C_6 = C_7 = 20\,m$ (full chord)

C_8 = final sub-chord $= 3.9\,m$.

Now, Calculation of deflection or tangential angle (θ)-

$$\theta = \frac{90C}{\pi R} \text{ (in degree)}$$

Where, C = Chord length in m.

R = Radius of curve (m)

Now,

θ for initial sub-chord ($C_1 = 7m$)

$$\theta_1 = \frac{90C_1}{\pi R} = \frac{90\times 7}{\pi \times 250} = 0°48'8''$$

Then, θ for all chords ($C_2 = C_3 = C_4 = C_5 = C_6 = C_7 = 20m$)

$$\theta_2 = \theta_3 = \theta_4 = \theta_5 = \theta_6 = \theta_7$$

$$= \frac{90\times 20}{\pi \times 250} = 2°17'31''.$$

Therefore, final sub-chord ($C_8 = 3.9\,m$)

$$\theta_8 = \frac{90\times 3.9}{\pi\times 250} = 0°26'49''.$$

Then Arithmetic check →

$$\theta/2 = \theta_1 + \theta_2 + \theta_3 + \theta_4 + \theta_5 + \theta_6 + \theta_7 + \theta_8.$$

$L.H.S = \theta/2 = 30/2 = 15°$ [∵ $\theta = 30°$, already calculated]

$R.H.S = \theta_1+\theta_2+\theta_3+\theta_4+\theta_5+\theta_6+\theta_7+\theta_8 = 0°48'8'' + 6(2°17'31'') + 0°26'49'' = 15°$

∵ LHS = RHS, So arithmetic check found correct.

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Station	Chainage (m)	Chord Length (m)	Tangential angle for each chord (Deg)	Deflection angle (deg)	Remark.
T_1	1183	—	—	—	Initial point of curve.
A	1190	7	0°48'8"	0°48'8"	Normal Chord.
B	1210	20	2°17'31"	3°5'39"	Normal Chord
C	1230	20	2°17'31"	5°23'10"	Normal chord.
D	1250	20	2°17'31"	7°40'41"	Normal Chord
E	1270	20	2°17'31"	9°58'12"	Normal Chord.
F	1290	20	2°17'31"	12°15'43"	Normal Chord
G	1310	20	2°17'31"	14°33'14"	Normal chord.
T_2	1313·9	3·9	0°26'49"	15°0'3"	Final point of curve.

Teacher's Signature:

Experiment No: 05.

Aim: Study of Prismatic compass.

Objective: To identify different parts of prismatic compass and to know their functions.

Terminology:

- **Compass box:** It is a circular box of diameter 85 to 110 mm having pivot at the centre and covered with plain glass at top.
- **Magnetic needle:** It facilitates in taking the bearings of survey lines with reference to the magnetic north.
- **Graduated ring:** The bearings are marked inverted on the graduated rings from 0° to 360° in clockwise starting 0° from south.
- **Pivot:** Magnet is freely held with this.
- **Object vane:** It consists of prism with a sighting slit at the top. The prism magnifies and erects the inverted graduations.

• Brake pin : It is pressed to stop the oscillations of the graduated ring.

• Lifting pin : On pressing it brings the lifting lever into action.

• Colour glasses : Red and blue glasses are provided with the prism to light luminous objects.

Instruments :
- Prismatic compass with tripod.
- Ranging rod.

Method of using Prismatic compass : The compass may be held in the hand, but for better results, it is usually mounted on a tripod. which carries a vertical spindle in a ball and socket joint to which the box is screwed. By means of this arrangement the instrument can be quickly levelled and also rotated in a horizontal plane and clamped in any position.

Working of Prismatic compass : This can be used while it in hand, but for better accuracy, it is usually mounted on a light tripod which carries a vertical spindle in the ball and socket arrangement to which compass is screwed. By means of this arrangement the compass can be placed in position easily. Its working involved the following steps:-

(i) Centering: The compass should be centered over the station where the bearing is to be taken by dropping a small piece of stone so that it falls on the top of the peg marking the Station.

(ii) Levelling: The compass should then be levelled by eye, by means of a ball and socket joint so that the ring may swing quite freely. It should be clamped when levelled.

(iii) Observing the bearing: To observe the bearing of a line AB.

(a) Centre the compass over the station A and level it.
(b) Having turned up vertical prism and the sighting vane, raise or lower ~~the~~ the prism untill the graduations are clearly visible.
(c) Turn the compass box untill the ranging rod at the station B is bisected by the hair when looked through the slit above the prism.

(d) When the needle comes, look through the prism and note the reading at which hair line produced appears to cut the image of the graduated ring which gives the required bearing of the line AB. Readings are usually estimated to the nearest 15'.

Procedure :

(1). Four ranging rods are fixed at different points (A, B, C, D, E) etc. such that it should be mutually visible and may be measured easily.

(2) Measure the distance between them. At point A the prismatic compass is set on the tripod stand, centering and levelling is then properly done.

(3) The ranging rod at B is ranged through sighting slits and objective vane attached with horse hair and reading on prismatic compass is noted down.

(4) It is forebearing of line AB. Then the prismatic compass is fixed at B and ranging rod at C and A is sighted. And reading is taken as forebearing of BC and back bearing of AB.

(5) Repeat the same procedure at the stations C, D, etc.

Observation Table:

Line	Distance (m)	Fore Bearing.	Back Bearing.
AB	5	52°30′	232°30′ (234°)
BC	5-2	79°	259° (258°)
CD	5-11	98°	278° (277°30′)

For Back Bearing Calculation:

- If FB is less than 180° → then, BB = FB + 180°
- If FB is more than 180° → then, BB = FB − 180°.